# Isaac Mubiru

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## PROFESSIONAL SUMMARY

Backend engineer with **6+ years running Node.js in production** for fully remote, async teams in the US, Singapore and Uganda. I own projects **end to end**: at MTailor I led the **zero-downtime migration of 20+ production applications** to Firebase/GCP, reporting to the CTO, including **moving file storage from Amazon S3 to Google Cloud Storage** and shutting down all AWS services for $5,000/month in savings. I have built **distributed systems** on Kafka, Pub/Sub, gRPC, Docker and Kubernetes, shipped **video processing with ffmpeg**, and written SDK documentation so other engineers could self-serve on a new platform. I have spent years building on Firebase, the platform Supabase is the open-source alternative to.

## CORE COMPETENCIES

Node.js in Production (6+ yrs) | Scalable Distributed Systems | Object Storage Migration (S3 → GCS) | Media Processing (ffmpeg) | Deploying & Scaling Services (Docker, Kubernetes, GCP) | Event-Driven Systems (Pub/Sub, Kafka, gRPC) | Developer Experience & SDK Docs | End-to-End Ownership

## WORK EXPERIENCE

### MTailor

Sep 2022 – Present

Software Engineer · US Remote · e-commerce

- Owned the end-to-end migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to the CTO.
- Implemented **real-time two-way sync** between MongoDB and Firestore in **Node.js** on Google **Pub/Sub**, keeping two live production databases consistent throughout the migration.
- Migrated file storage from **Amazon S3 to Google Cloud Storage** with a Python script, and moved all remaining services off AWS (S3, EC2, EBS), saving **$5,000/month**.
- Built a 3D visualisation feature using **video overlay and ffmpeg** that increased buyer conversion; shipped Express Shipping (**+$40 revenue per order**) and migrated a Webflow site from Parse to Firebase.

### MTailor

Mar 2022 – Jul 2022

Backend Contractor · US Remote

- Built the migration script that read from MongoDB, ran complex processing logic and wrote to Firestore.
- Wrote **developer documentation** for fellow engineers on the new Firebase SDKs (Node.js and Python) and trained the Ops team on the new dashboard and backup procedures.

### Dr Wealth

Aug 2021 – Jan 2022

Software Engineer · Singapore Remote · fintech

- Extended backend APIs in **TypeScript** on Firebase Cloud Functions and Express.js (Heroku) for a real-time stock-price PWA.
- Built ad-hoc jobs over **2M+ Firestore records** to keep customer prices current from Morningstar APIs; added responsive React/Tailwind pages.

### Mind2matter

Jan 2022 – Jul 2022

Software Engineer · US Remote · agency

- Built Node.js backends for client applications and delivered React UIs from Figma designs under tight timelines, reporting to the CTO.

### CodeBits

Jan 2020 – Jul 2021

Software Engineer / Team Lead · Uganda Remote

- Architected a **microservices** backend for FIDA Uganda's case management app using **Apache Kafka, Docker and Kubernetes**, with NATS Streaming and gRPC between services.
- Led a team of 4 developers building legal-tech systems for NGOs, including a paralegal database for LASPNET Uganda.

- Built a USSD service for legal-aid providers that talks to a Node.js backend via **gRPC**, and a React Native + Expo app with Firebase push notifications.

## PROJECTS

### Real-time Pub/Sub Sync Service

distributed systems

Two-way MongoDB↔Firestore sync running under parallel production traffic across 20+ applications during a zero-downtime migration.

Node.js · Google Pub/Sub · MongoDB · Firestore

### S3 → GCS Storage Migration

object storage

Moved MTailor's stored files from Amazon S3 to Google Cloud Storage as part of the AWS exit that cut $5,000/month in infrastructure cost.

Python · Amazon S3 · Google Cloud Storage

## EDUCATION

**B.Sc. Computer Engineering** — Makerere University 2018 – 2022

Kampala, Uganda

## SKILLS

**Languages:** JavaScript, TypeScript, Python, SQL, Go (intermediate), Rust (intermediate)

**Backend:** Node.js, Express.js, REST APIs, gRPC, Microservices, Serverless (Firebase Cloud Functions)

**Data & Storage:** MongoDB, Firestore, Google Cloud Storage, Amazon S3, Google Pub/Sub, Apache Kafka, NATS Streaming

**Cloud / Infra:** GCP, Firebase, AWS (S3, EC2, EBS — intermediate), Docker, Kubernetes, Git, Linux

**Media / Frontend:** ffmpeg, React, React Native, Tailwind, HTML, CSS **Spoken:** English, Kiswahili